

SHA-1 Message Digest Algorithm

Aim:

To write a C++ program to calculate to 160-bit message digest using SHA-1 and verify message integrity across client-server connection.

Algorithm:

Client Processing:

* Compute SHA-1 digest using

`EVP_DigestInit_ex(EVP_sha1())`

* Convert digest to a 40-character hexadecimal string

* Send plaintext and digest to server.

Server Verification

* Receive plaintext and digest.

* Recompute SHA-1 locally.

* Compare digests for integrity verification.

Output:

Client:

Message: "The quick brown fox jumps over the lazy dog"

SHA-1 Digest: 2fd4e1c67a2d28fc ed149cc1bb76c7-
391b93cb12

sent message and sha-1 digest successfully.

Server:

Listening on port 8080...

Received plaintext message: ~~The fox jumps~~

"The quick brown fox jumps over the lazy dog"

Received SHA-1 Digest: 2fd4e1c67a2d28fced849cc1bb76c7391b93cb12

Computed SHA-1 Digest: 2fd4e1c67a2d28fced849cc1bb76c7391b93cb12

~~Integrity~~ Integrity verified: Digest matches perfectly!